

Kaduna.lk

A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

PP2 Feedback and Actions

Feedback	Action
Module accuracy and effectiveness should be validated with measurable evidence, supported by stakeholder feedback demonstrating practical relevance, usability, and reliability	• Validated module accuracy and effectiveness through measurable evaluations. • Conducted real-world testing with industry experts. • Collected expert feedback to assess practical relevance. • Confirmed the solution's usability and reliability through expert feedback.



Automobile Association of Ceylon (AAC)



Allianz Finance





Garage Visits



Department of Motor Traffic (DMT)

The Problem

Sri Lanka has **NO dedicated digital platform** for roadside assistance



Stranded drivers have no help



No dispatch prioritization or congestion assessment



No proper approach for vehicle Accident claims



No proper approach for the vehicle maintenance

Solution Overview

Kaduna is an smart roadside assistance platform designed to provide fast, accurate, and intelligent support for stranded drivers while improving dispatch efficiency, reducing traffic impact, and enabling proactive vehicle maintenance.

Key Feature

1. Smart Dispatch & Recovery
2. Geo-Intelligence & Traffic Impact
3. Predictive Maintenance
4. Intelligent 3D Accident Claims
5. User-Friendly Mobile Platform







Smart Dispatch & Recovery



Driver input

Adaptive questions: dashboard lights, starting sounds, smells, engine state. OBD-II readings.

1 · Diagnostic triage

Decision tree outputs a probability for each of 29 faults, plus an entropy score.

2 · ECM optimizer

Expected cost per provider: travel + service + mismatch risk, with traffic and trust. Picks the lowest.

Provider dispatched

Ranked list, with every cost breakdown saved for audit.

3 · Bayesian feedback loop

When a job closes, the confirmed fault updates a discounted Dirichlet posterior for that symptom pattern ($\gamma = 0.99$). After 3 or more outcomes it is blended with the tree, so the next similar case is diagnosed better.

Smart recovery

If a provider declines or stays silent for 120 s, a watchdog (polling every 15 s) re-runs ECM without the providers already tried. A hard capability gate stops a mechanic from outranking a tow truck for a tow-only job.

Smart Dispatch & Recovery

Results Acuracy

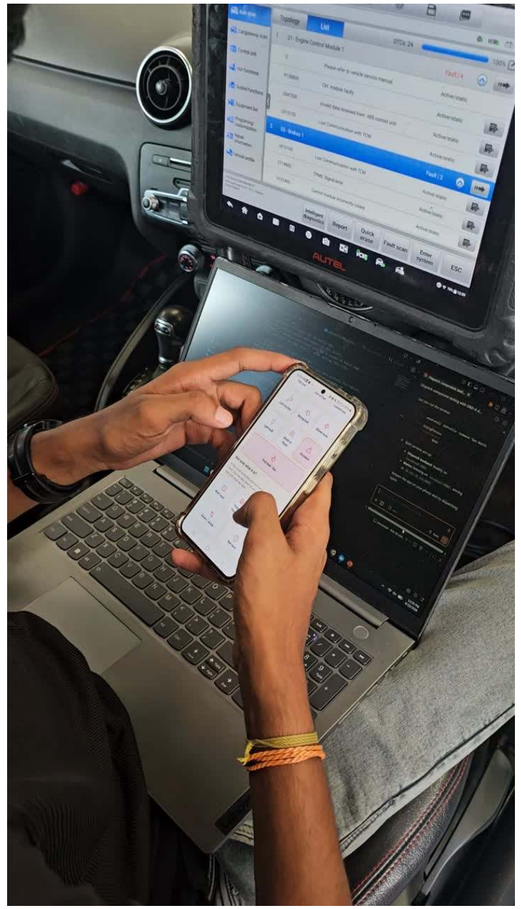
Tier	Model	Accuracy	Macro-F1	ECE	Depth
Q Only	DT	82%	0.70	22.50	19/213
Q Only	RF	83.20%	0.78	19.20	200 trees
Q + OBD	DT	87.80%	0.84	25.60	17/149
Q + OBD	RF	88.60%	0.86	27.50	200 trees

Observed Evidence

Incidents	242
Saved Diagnosis	140
Incident Ranking Rcords	115
Resolution feedback records	10

ML triage (battery, radiator fan)	5	**5/5**
Fast path (flat tyre)	5	**5/5**

- Also 97% provider dispatched in Tree + ECM, simulation, 30 seeds × 1,000 incidents
- 26% - 50.1% Battery_Replace in 31 repeated pattern



Expert feedbacks



Mr. Devapriya Hettiarachchi
Secretary - AACL



Mr. Dileepai Wari
Automobile Mechanic
Motorsport Expert - Lap 57



Mr. Dilanka Rajapaksha
Automobile Expert
Aida Automotorist

SLIIT FACULTY OF COMPUTING

Geo-Intelligence & Traffic Impact



Incident reported

GPS position, time, incident type and lanes blocked, sent by dispatch for every help request.

1 · Road lookup

Road class and lanes read from the real Colombo OpenStreetMap network at the incident's GPS point.

2 · Impact score

Lanes blocked 25%, traffic volume 25%, time 20%, road type 15%, incident type 15%. Score 1–10, LOW to CRITICAL.

Score to dispatch

Returned in under 10 ms with the factor breakdown. If geo is down, dispatch uses a neutral 5.

3 · Impact hotspots

KDE + DBSCAN (0.5 km) over 500 incidents on the real Colombo network found 25 hotspots. High-impact incidents are 2.3× more likely to fall inside one; 10 of 25 sit near police blackspots.

Validation

120 SUMO traffic simulations: Spearman 0.67 (target 0.75 missed); lanes blocked drives delay most (r 0.88). Impact-first dispatch was tested and rejected, so the score is used for triage. DMT expert SUS 80.

Geo-Intelligence & Traffic Impact



Results

Test	Measure	Result	Target	Outcome
Score vs 120 SUMO runs	Spearman (Pearson r)	0.67 (0.60)	0.75	Missed
Impact hotspots (25)	Silhouette · odds ratio	0.64 · 2.3	> 0.3	Met
Impact-first dispatch (H2)	Delay vs nearest-first	19.1% worse	Lower delay	Rejected
Live scoring API	p95 latency (146 tests)	< 10 ms	500 ms	Met

Expert feedbacks



Mr. Chamila Wasana Muthukumarana
Assistant Commissioner, DMT
SUS 80 · relevance 4.4 / 5



Mr. Sarath Kumara
Ex-DIG, Sri Lanka Police
Found the road-class bug (fixed)



AAC Engineering Unit
Automobile Association of Ceylon
UMUX-Lite 71.7 · triage 3 / 3

Predictive Maintenance



OBD-II connection

Pair an ELM327 adapter via Bluetooth. Read supported engine signals and diagnostic trouble codes.

1 · Trip features

Combine OBD-II readings with phone motion sensors to measure braking, cornering and vehicle usage.

2 · RUL prediction

Random Forest and XGBoost estimate remaining kilometres for the engine, brake pads, tyres and battery.

Maintenance plan

Show component health and service urgency. Recommend suitable parts and nearby garages.

3 · Expert suggestions

Maintenance rules set urgency and actions. AI explains recommended work using curated automotive references, with source citations when available.

Service history updates

Track the odometer and component replacements. Reset the replaced part's health and use its mileage history in later assessments.

Predictive Maintenance

Validations

Expert feedbacks

Simulation-Based Scenario Testing

Accuracy of ml models

Component	R^2	MAE (km)	RMSE (km)
Engine	0.8935	11,791.30	15,054.20
Brake pads	0.7292	4,754.60	6,971.10
Tyres	0.7088	5,939.70	8,105.20
Battery	0.9201	5,212.90	7,226.70



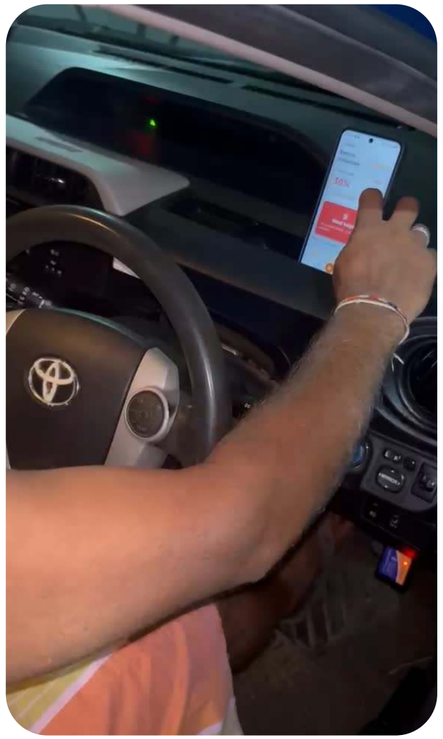
Mr. Devapriya Hettiarachchi
Secretary - AACL



Dinesh Ayeshmantha
Ayesh Automobile (Mechanic)
17 years



Vihanga Wedage
Vehicle Technical & Valuation Officer
Peoples' Finance



Intelligent 3D Accident Claims



Guided capture

Smartphone photos in a 3-height ring (overhead, chest, waist), uploaded to Cloudflare R2. The insurer starts the 3D pipeline from the dashboard.

1 · Brightness check

Mean brightness picks a tier: Zero-DCE (50–60), gamma (60–100) or none (≥100). Photos 20% darker than the rest are normalised.

2 · COLMAP (SfM)

SIFT features matched across photos. Bundle adjustment solves every camera pose and builds a sparse 3D point cloud.

3 · Gaussian Splatting

nerfstudio splatfacto trains ~700k 3D Gaussians from the poses, on a serverless Modal NVIDIA L4 GPU.

3D model in the dashboard

The model is exported as splat.ply to R2 and shown in a WebGL viewer. The insurer can rotate, pan and zoom around the damaged car with no plugins. Live progress is polled every 4 s.

Low-light fallback

If mean brightness is below 50, Zero-DCE, a shadow lift and YCbCr denoising brighten every photo. The enhanced photos go to the dashboard and 3D is skipped.

Intelligent 3D Accident Claims



Time Taken for Guided Capture 10-12 min

Results Acuracy

4 Key Scenarios **Scenarios - 15** **Models Generated - 60+**

Test Scenario	PSNR (dB) ↑	SSIM ↑	LPIPS ↓	Reproj. error (px) ↓	Matches per pair ↑	BRISQUE ↓	Time
Outdoor (3-height ring)	27.58	0.91	0.12	1.23	394	24.30	23.5 min
Outdoor (single photo)	27.5	0.83	0.22	1.18	585	28.69	19.2 min
Indoor garage	25.36	0.74	0.24	1.29	509	26.76	27.8 min
Low light (enhanced)	–	–	–	–		47.81 → 43.22	4.7 min

Expert feedbacks



Dr Ajith Raveendra De Mel
Chairman of IRCSL
Insurance Regulatory Commission of Sri Lanka



Mr. Lahiru Senadeera
Relationship Manager
Allianz Insurance
Sri Lanka



Mr. L. K. W. P Perera
Chief Executive Officer (CEO)
Sanasa Insurance
Sri Lanka



Mr. Indika Kiriwandeniya
Managing Director (MD)
Sanasa Insurance
Sri Lanka

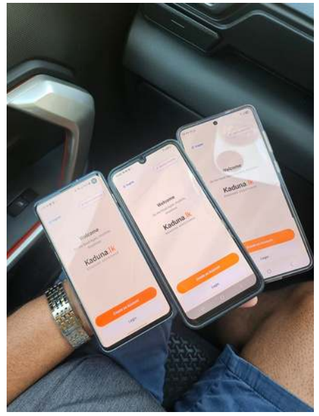
Testimonials

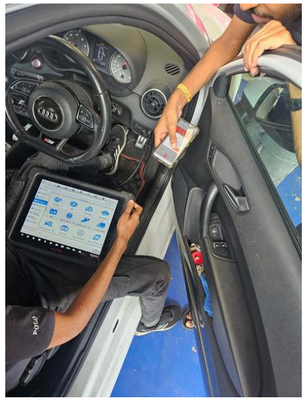
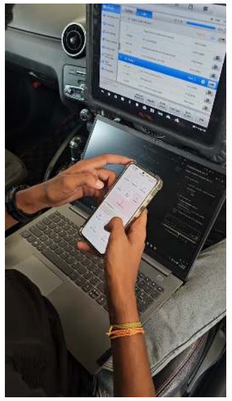


Stakeholder Validations







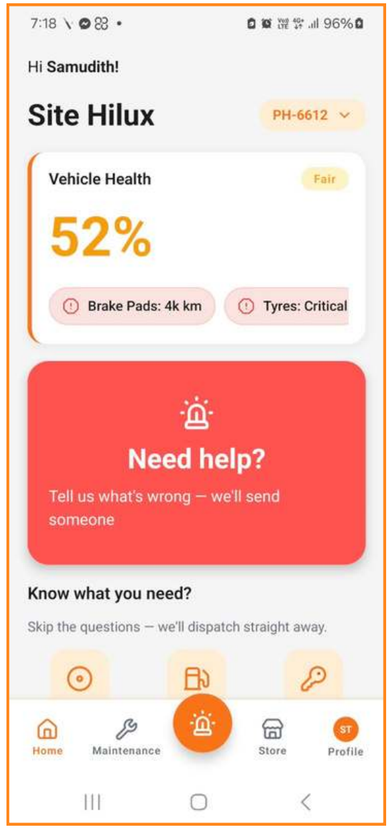










User Reqested UX Enhancements

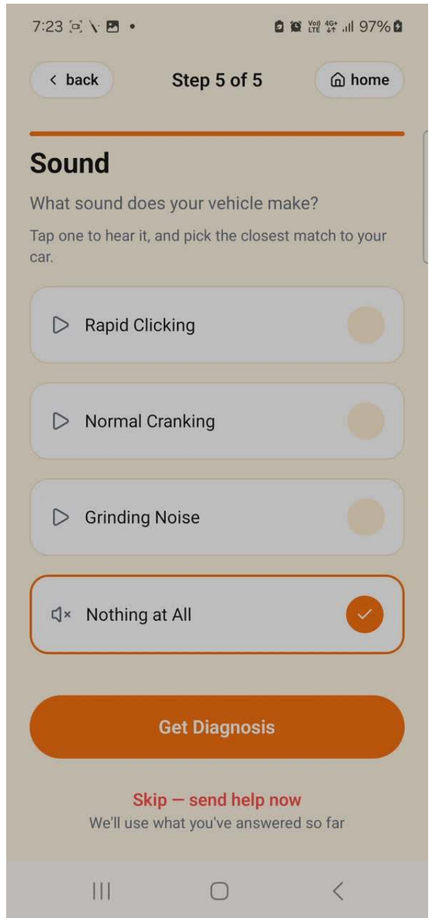




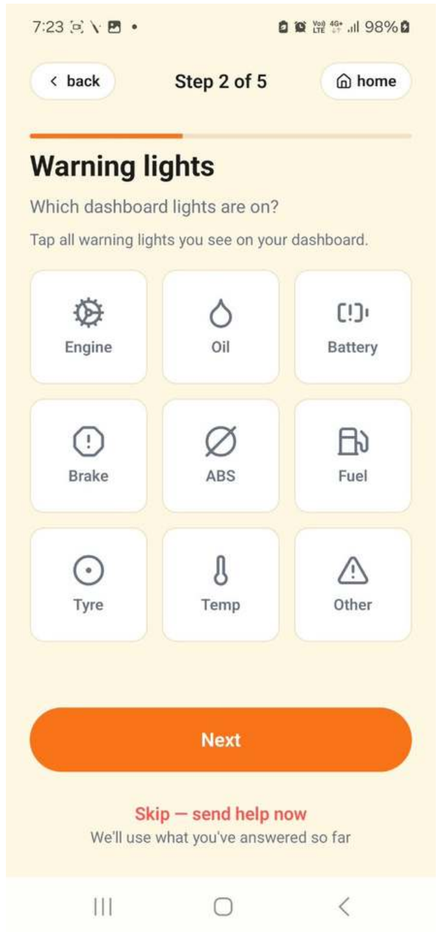


Trilingual Interfaces

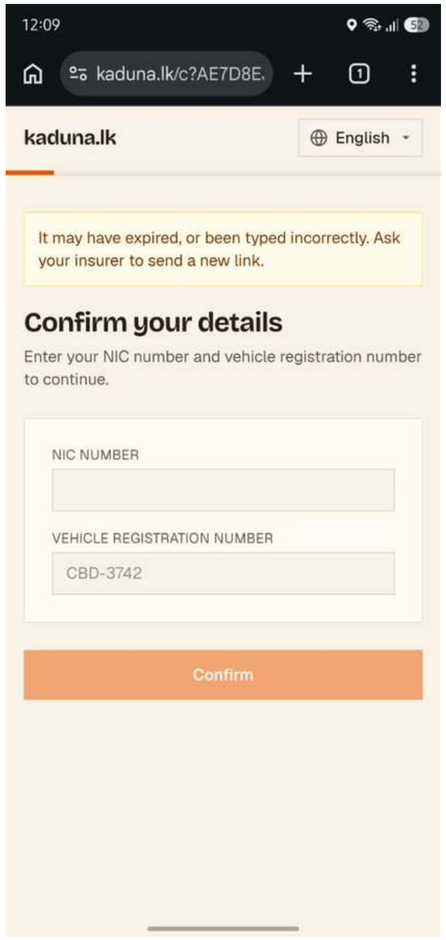
User Reqested UX Enhancements



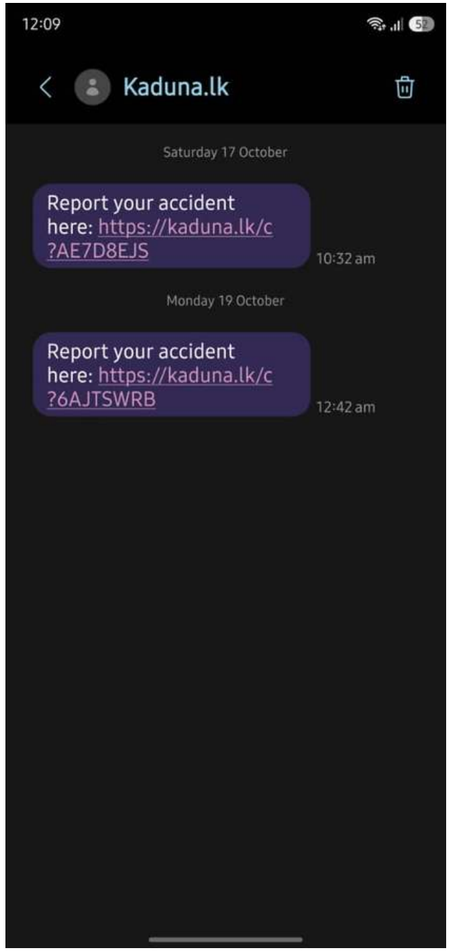
Starting Sounds



Dashboard Lights



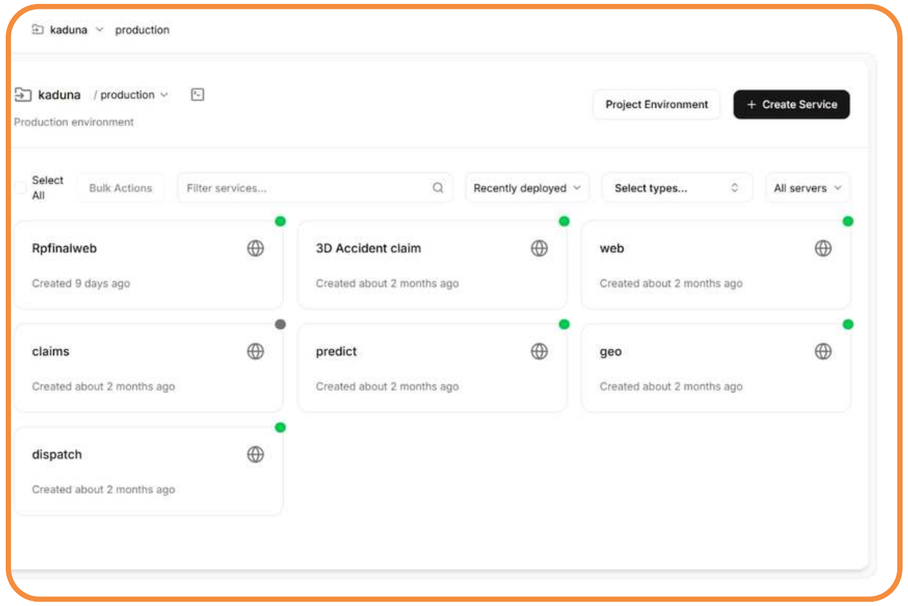
Web insurance claim



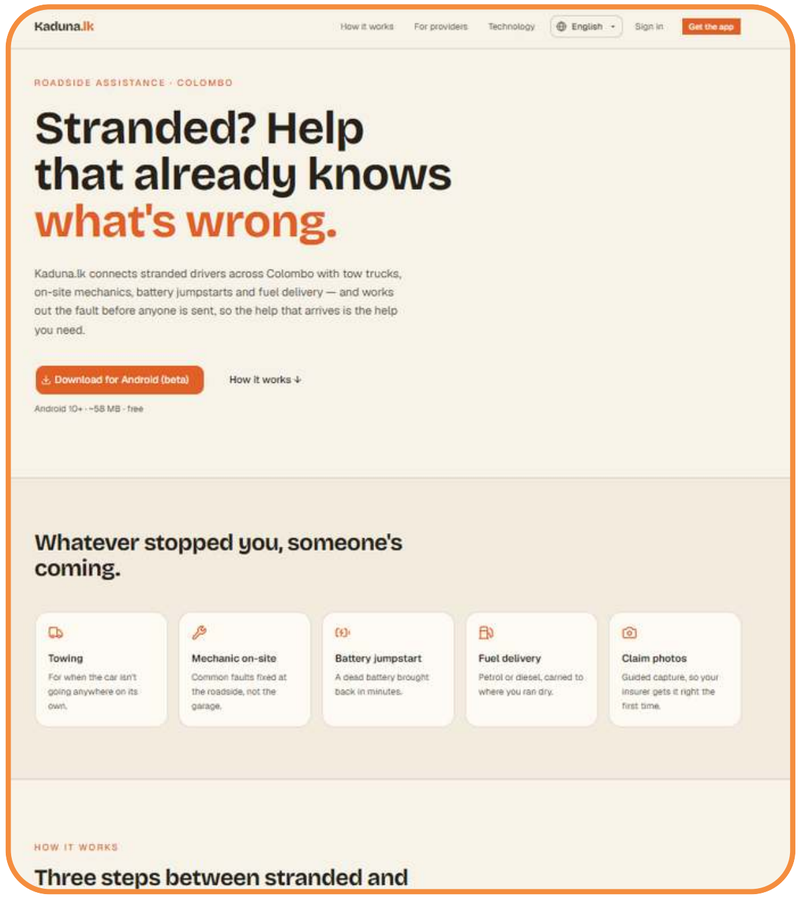
Offline messaging

Real World Readiness & Recognition

Deployed & Integrated

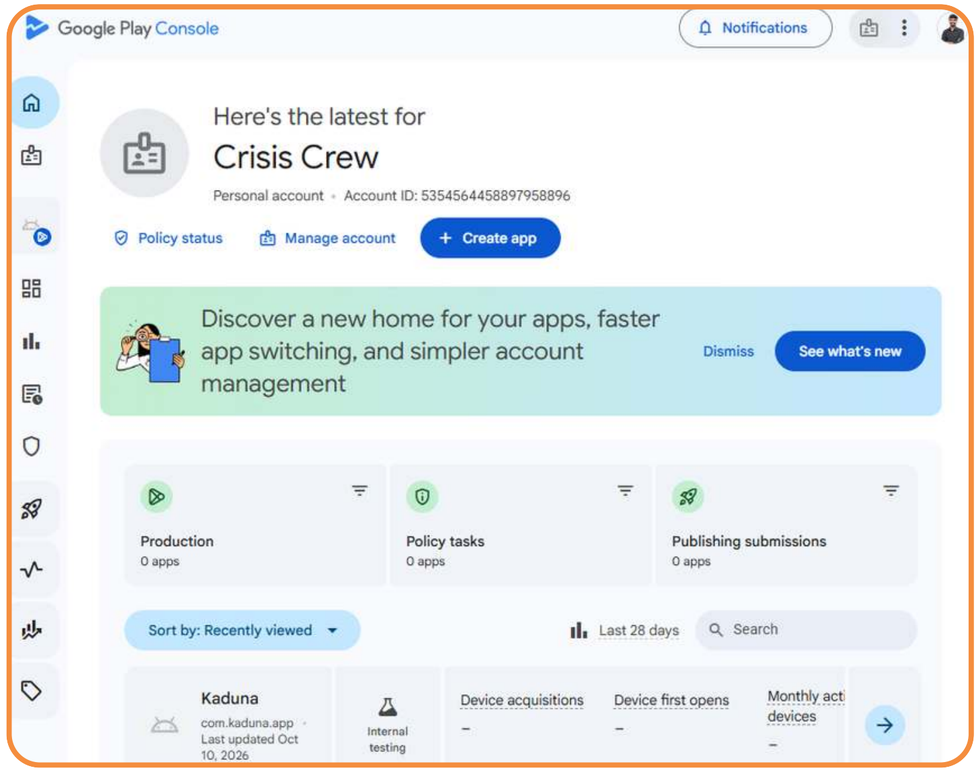


Live dispatch service on the team VPS, push-to-deploy, used by the Kaduna.lk mobile app and dashboard.



Real World Readiness & Recognition

Play Store Testing





Android app (com.kaduna.app) is on a Google Play testing track with 12 real testers.

Real World Readiness & Recognition

ICCMM 2026

An abstract of this research has been accepted at ICCMM 26 to present at the conference held on 12th- 14th November.



ICCMM 2026
International Conference on Computational and Mathematical Modelling

In Collaboration With
33rd FAI International Conference

21st September 2026

Mr. Janukshan Sivakumar,
Department of Software Engineering,
Sri Lanka Institute of Information Technology,
Malabe,
Sri Lanka.

Dear Mr. Janukshan Sivakumar,

Acceptance of Abstract for Presentation at ICCMM 2026

We are pleased to inform you that, after being considered by the conference review team, your abstract, titled **"UADO: Uncertainty-Aware Dispatch Optimization for Roadside Assistance via Expected-Cost Minimization and Bayesian Feedback"** has been accepted for oral presentation at the **International Conference on Computational and Mathematical Modelling 2026 (ICCMM 2026),** hosted by the Centre for Mathematical Modelling, Faculty of Science, University of Colombo from 12th to 14th November, 2026.

Your paper tracking number is **ICCMM_26_462.** Kindly pay the registration fee before the deadline using the details provided below.

Bank Details

- Beneficiary Name: Colombo Science and Technology Cell
- Beneficiary Account: 1116016487
- Bank: Commercial Bank of Ceylon PLC
- Branch: Reid Avenue
- Bank Routing Sort Code and/or Branch Code: 7056-116
- SWIFT/BIC Code: CCEYLKLX
- Account Currency: LKR
- Bank Country: Sri Lanka

* Please note that the registration fee does not include the accommodation facilities.

We look forward to your active participation in making ICCMM 2026 a success.

Best regards,

Prof. Sanjeewa Perera
Conference Chair
ICCMM 2026

Dr. Hasitha Erandi
Conference Secretary
ICCMM 2026

* Please visit https://iccmm.cmb.ac.lk/2026/#registration for more registration details.

Centre for Mathematical Modelling
Department of Mathematics
Faculty of Science
University of Colombo

Website: https://iccmm.cmb.ac.lk/2026
E-mail: cmm@maths.cmb.ac.lk

Prof. Sanjeewa Perera +94 77 775 1627 (WhatsApp Only)
Dr. Hasitha Erandi +94 77 288 9866 (WhatsApp Only)

Real World Readiness & Recognition

Hack-X 11.0 Finalists

Selected to the Grand Finale to be held on 11th of Novemeber of Inter-University Hackathon conducted by University of Kelaniya representing SLIIT



Commercialization

8M+
Registered Vehicles in Srilanka

1200 - 1500
Insurance Approved Garages

60 - 80
Service Center Franchise

*According to the **Sri Lanka Automobile Service Providers Association (SLASPA)** and the Department of Motor Traffic (DMT)*

Free Plan
Basic vehicle assistance features are available at no cost to drivers.

Premium Plan
Advanced features and benefits for power users and families.

B2B Subscription
Recurring subscription plans for fleet operators and insurance companies.

Service Booking Fees
Earn a commission on each booking made through the platform with garages and service centers.

Marketplace Commission
Earn a percentage from every product or service sold through the marketplace.

Advertising
Generate revenue from targeted promotions by parts suppliers and service brands.

Phase 01 Implementation

Automobile Association of Ceylon (AAC)

LAP 57 Boralegsgamuwa

Allianz Insurance Lanka

Department of Motor Traffic

Sanasa Insurance

Volmec Svenska Bilar Malabe

Aida Motorist Dehiwala

Limitations & Future Improvements

Limitations

29 Fault Classes

Currently the system diagnoises 29 fault classes

A noisy reading can cause a lasting low score in RUL prediction

Next steps

Expand Diagnoise Triage with more fault classes

Add more fault classes to expand the diagnosis range and to make the system more efficient

Demo roadmap

1. Introduction to the Application
2. Fault Diagnosis and Service Provider Dispatching
3. Traffic Impact Scoring, Incident Hotspot Dashboard
4. Prioritize Hotspots and Locations
5. OBD Telemetry Integration, Predictive Maintenance
6. AI-Based Expert Support
7. 3D Model Demonstration, Lowlight enhancement
8. How the Insurance Officers will use the system

Thank You !

Kaduna.lk

Questions & Discussion